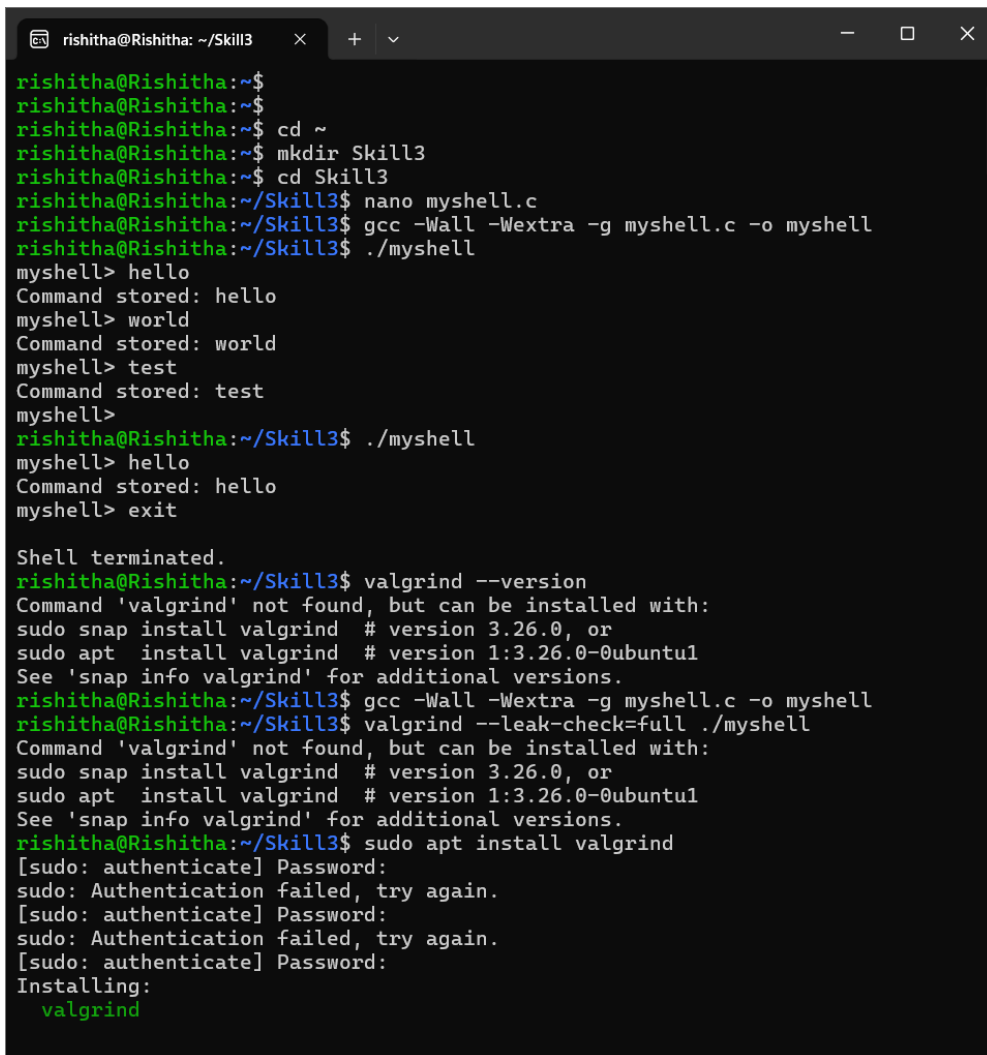


# SKILL-3

1. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.



```
rishitha@Rishitha: ~/Skill3
rishitha@Rishitha:~$
rishitha@Rishitha:~$
rishitha@Rishitha:~$ cd ~
rishitha@Rishitha:~$ mkdir Skill3
rishitha@Rishitha:~$ cd Skill3
rishitha@Rishitha:~/Skill3$ nano myshell.c
rishitha@Rishitha:~/Skill3$ gcc -Wall -Wextra -g myshell.c -o myshell
rishitha@Rishitha:~/Skill3$ ./myshell
myshell> hello
Command stored: hello
myshell> world
Command stored: world
myshell> test
Command stored: test
myshell>
rishitha@Rishitha:~/Skill3$ ./myshell
myshell> hello
Command stored: hello
myshell> exit

Shell terminated.
rishitha@Rishitha:~/Skill3$ valgrind --version
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
rishitha@Rishitha:~/Skill3$ gcc -Wall -Wextra -g myshell.c -o myshell
rishitha@Rishitha:~/Skill3$ valgrind --leak-check=full ./myshell
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
rishitha@Rishitha:~/Skill3$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Installing:
  valgrind
```

## Installing dependencies:

|                             |                                   |                                         |
|-----------------------------|-----------------------------------|-----------------------------------------|
| <code>gdb</code>            | <code>libdebuginfod-common</code> | <code>libipt2</code>                    |
| <code>libbabeltrace1</code> | <code>libdebuginfod1t64</code>    | <code>libsource-highlight-common</code> |
| <code>libc6-dbg</code>      | <code>libdw1t64</code>            | <code>libsource-highlight4t64</code>    |

## Suggested packages:

`gdb-doc gdbserver valgrind-mpi kcachegrind`

## Summary:

Upgrading: 6, Installing: 10, Removing: 0, Not Upgrading: 108  
Download size: 37.8 MB  
Space needed: 111 MB / 1024 GB available

Continue? [Y/n] y

Get:1 <http://archive.ubuntu.com/ubuntu> resolute-updates/main amd64 libc-dev-bin amd64 2.43-2ubuntu2.4 [23.3 kB]Get:2 <http://archive.ubuntu.com/ubuntu> resolute-updates/main amd64 libc6-dev amd64 2.43-2ubuntu2.4 [2294 kB]Get:3 <http://archive.ubuntu.com/ubuntu> resolute-updates/main amd64 libc-gconv-modules-extra amd64 2.43-2ubuntu2.4 [1362 kB]Get:4 <http://archive.ubuntu.com/ubuntu> resolute-updates/main amd64 libc6 amd64 2.43-2ubuntu2.4 [2104 kB]Get:5 <http://archive.ubuntu.com/ubuntu> resolute-updates/main amd64 libc-bin amd64 2.43-2ubuntu2.4 [701 kB]Get:6 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]Get:7 <http://archive.ubuntu.com/ubuntu> resolute-updates/main amd64 locales all 2.43-2ubuntu2.4 [4225 kB]Get:8 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]Get:9 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]Get:10 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]Get:11 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]Get:12 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]Get:13 <http://archive.ubuntu.com/ubuntu> resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]

rishitha@Rishitha: ~/Skill3

```
Preparing to unpack .../03-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../04-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../05-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../06-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../07-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../08-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../09-libc6-dbg_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../10-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up locales (2.43-2ubuntu2.4) ...
Generating locales (this might take a while)...
Generation complete.
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libc-dev-bin (2.43-2ubuntu2.4) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2.4) ...
Processing triggers for systemd (259.5-0ubuntu3.4) ...
```

```
Processing triggers for systemd (259.5-0ubuntu3.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
rishitha@Rishitha:~/Skill3$ valgrind --version
valgrind-3.26.0
rishitha@Rishitha:~/Skill3$ valgrind --leak-check=full ./myshell
==2046== Memcheck, a memory error detector
==2046== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2046== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2046== Command: ./myshell
==2046==
myshell> hello
Command stored: hello
myshell> world
Command stored: world
myshell> exit

Shell terminated.
==2046==
==2046== HEAP SUMMARY:
==2046==    in use at exit: 0 bytes in 0 blocks
==2046==   total heap usage: 6 allocs, 6 frees, 1,168 bytes allocated
==2046==
==2046== All heap blocks were freed -- no leaks are possible
==2046==
==2046== For lists of detected and suppressed errors, rerun with: -s
==2046== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
rishitha@Rishitha:~/Skill3$ |
```